

This ensures that the new embeddings and unembeddings maintain a scale consistent with the original vocabulary.

H EFFICIENCY GAINS FROM VOCABULARY EXPANSION

Beyond improving model performance on newly added words, our vocabulary expansion method directly reduces the number of tokens required to encode input text, and could lead to further potential efficiency gains in inference.

Table 5 summarizes the average reductions in sequence length achieved by encoding texts from each domain using the expanded vocabulary instead of the original vocabulary in the three datasets. Using our method, we attempt to expand Llama2-7B’s vocabulary with all multi-token words appearing at least m times in the test set, where we use $m = 1, 5, 50$ for WIKITEXT-103, PUBMED, and Arabic WIKI40B, respectively.

Dataset	# Attempted	# New Words	Token Reduction
WIKITEXT-103	14.1k	10.1k	10.5%
PUBMED	9.5k	5.4k	13.5%
WIKI40B-Arabic	4.4k	0.7k	14.5%

Table 5: Reduction in average sequence length when encoding text with the expanded vocabulary for Llama2-7B. *# Attempted* is the number of multi-token words tested for expansion, while *# New Words* is those detected as detokenized using Patchscopes.

We find that the token reduction rates depend on the success rate of extracting detokenized representations using Patchscopes (which in turn depend on the textual domain and language). WIKITEXT-103 achieves a 10.5% reduction in token count, with 72.9% of attempted words successfully converted into new vocabulary entries. PUBMED shows a higher token savings rate (13.5%) despite a lower success rate (58.3%), as its vocabulary expansion targets more domain-specific multi-token terms. The largest efficiency gain is observed in Arabic WIKI40B, where encoding with the expanded vocabulary reduces token count by 14.5%, highlighting the method’s potential for languages with inherently longer token sequences. However, the success rate of identifying detokenized representations in WIKI40B is lower (16.0%), suggesting room for improvement in expanding vocabularies for morphologically rich languages, and suggesting many Arabic words are not stored in the model’s inner lexicon.

These results suggest that post-hoc vocabulary expansion can significantly reduce the computational cost of inference, particularly for non-English languages and domain-specific texts, without requiring any modifications to the model’s core parameters.